

2 (**short**) The Coefficient of Performance of a heat pump, COP_P , is defined as

$$\text{COP}_P = \frac{\text{heat to hot space}}{\text{work in}}$$

(a) A domestic heat pump takes heat from the atmosphere at 5 Celsius. Calculate the maximum COP_P when:

- (i) maintaining water at 40 Celsius for central heating;
- (ii) maintaining water at 50 Celsius for washing;
- (iii) maintaining air at 250 Celsius for cooking in the oven.

Comment on your answers. [4]

(b) Calculate the minimum work required to heat 5 kg of water from 20 Celsius to 90 Celsius taking heat from the atmosphere at 10 Celsius. [6]